AUTOMATION CASE STUDY

Vendor COI chasing

Document intake, expiry tracking and approval-led vendor follow-up.

I built an automation that collects vendor certificates of insurance, extracts dates and coverage fields, maintains a vendor register and prepares follow-up requests for human approval.

Collect

Gmail and form submissions bring PDFs into one consistent intake process.

Track

Stored records connect original PDFs, expiry dates and review flags.

Follow up

A daily watch prepares requests and waits for a person to approve each send.

Business problem

Certificates arrive across emails and uploads, while expiry dates and missing documents require repeated manual checking. The build connects these documents to one register and an approval-led follow-up process.

Demonstration scope: synthetic vendors and certificate-like documents on a local n8n/Postgres stack. OpenAI extraction and drafting used the real service. Preserved Gmail runs delivered to a test mailbox. No mock application screens are presented as evidence.

ARCHITECTURE

Six workflows. One durable register.

n8n connects the process; Postgres holds certificate history, notices and outreach state.

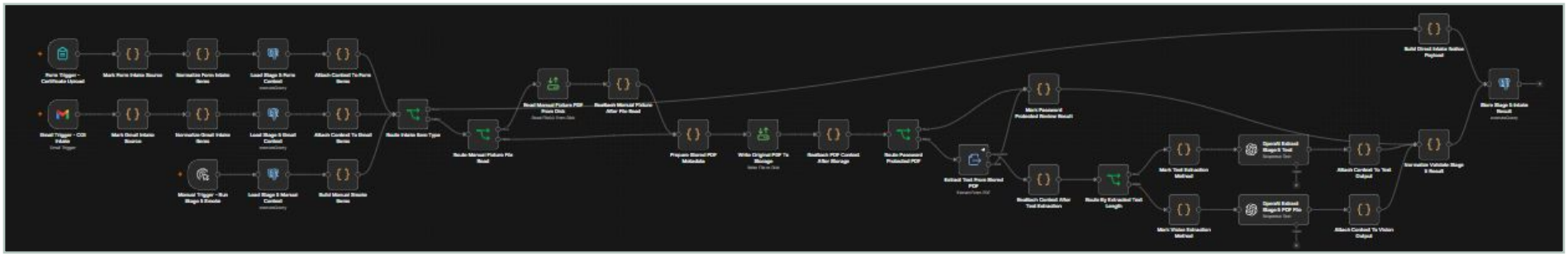
COI Intake	Gmail / form → PDF storage, extraction, checks → certificate or notice
Source Document Route	Dashboard record ID → stored source lookup → original PDF response
Daily Watch and Chase	Schedule → vendor/outreach state → drafts and pending outreach
Vendor Approval Request	One request → Gmail approval → sent or declined outreach
Non-PDF Reply Notice	Gmail message without PDF → normalization → durable notice
COI Dashboard	Browser request → Postgres register rows → refreshed dashboard

Core systems: n8n, Postgres, Gmail, OpenAI, the upload form and certificate file storage.

01 / COI INTAKE

From a PDF to a usable record

Full actual canvas after n8n Tidy Up. Readable branches follow.



Trigger and incoming data

Gmail attachments, the certificate upload form or a manual synthetic test enter the same intake pipeline. Each PDF retains its source context.

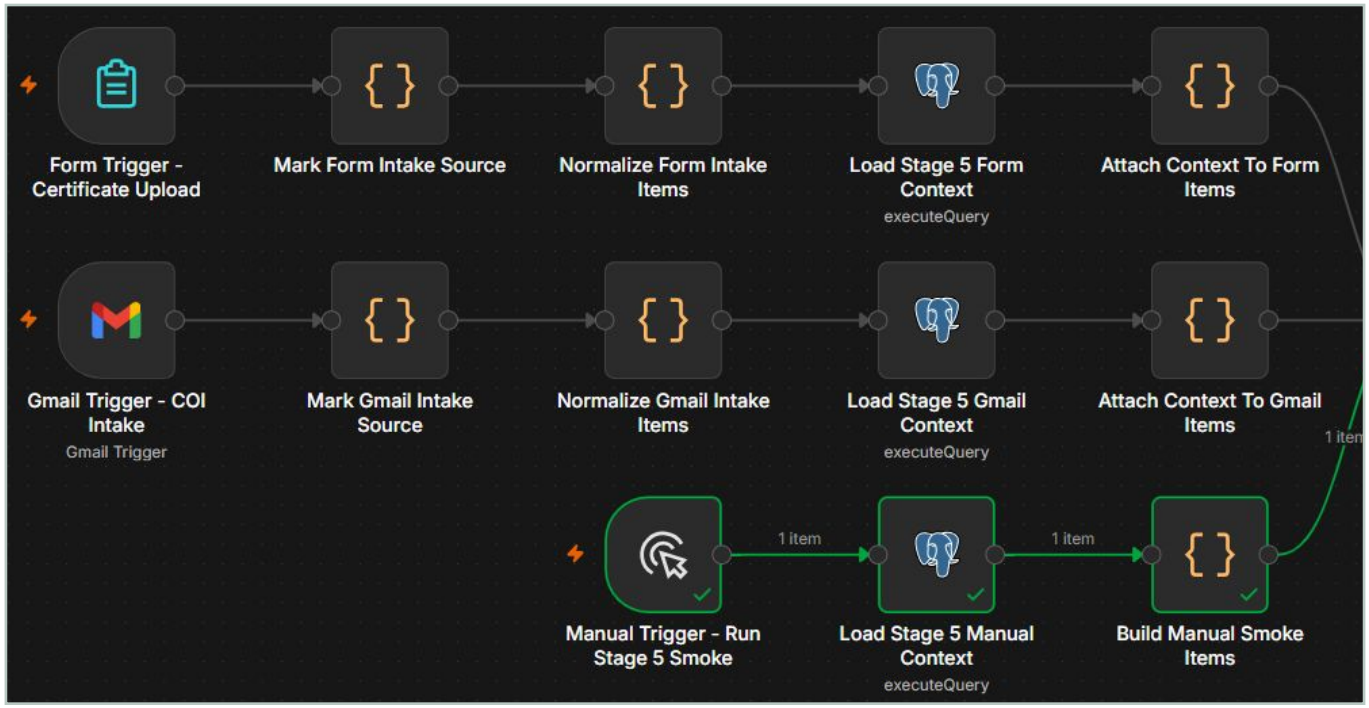
Output and human review

The original is saved, fields are extracted and deterministic checks choose a stored certificate or a notice. Ambiguous documents receive a review state.

01 / INTAKE ENTRY POINTS

Email and form inputs converge

Actual form and Gmail branches; the green manual path is the fresh synthetic run.



Normalize first

Source-specific steps normalize attachments and form fields, then load vendor and holder context from Postgres.

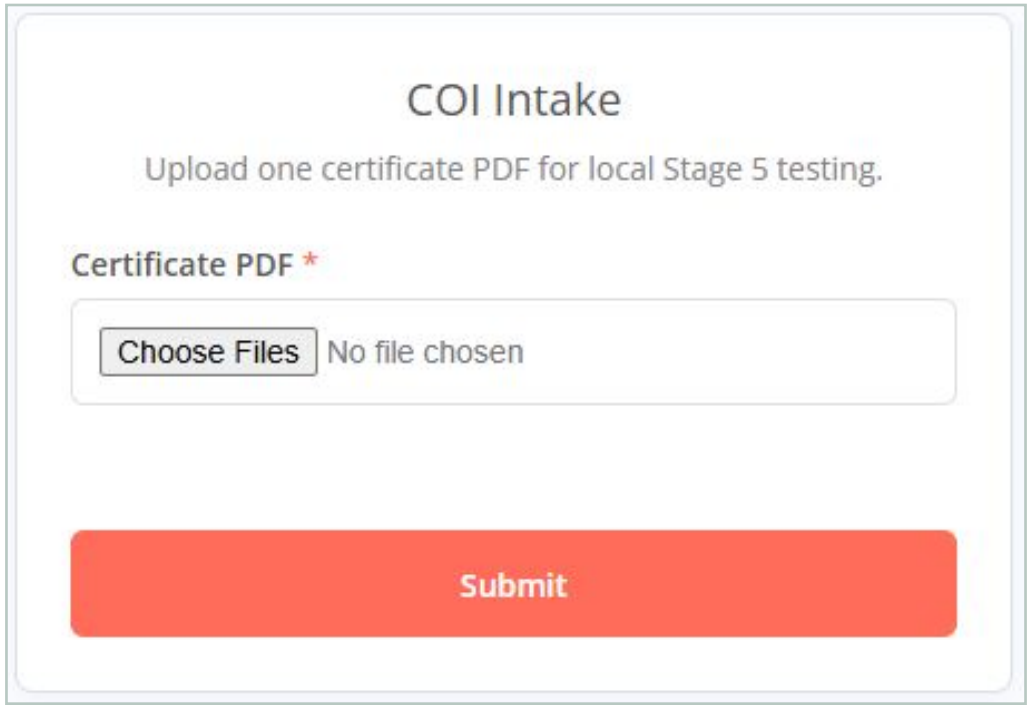
Connected evidence

The fresh run used Apex Plumbing Services LLC. A preserved form execution also stored three submitted PDFs; its results appear on page 8.

01 / UPLOAD INTERFACE

A simple certificate upload form

Actual n8n form served by the running application. This capture shows the entry interface.



Incoming data

A user submits a certificate PDF. The form trigger normalizes the upload and attaches the required context.

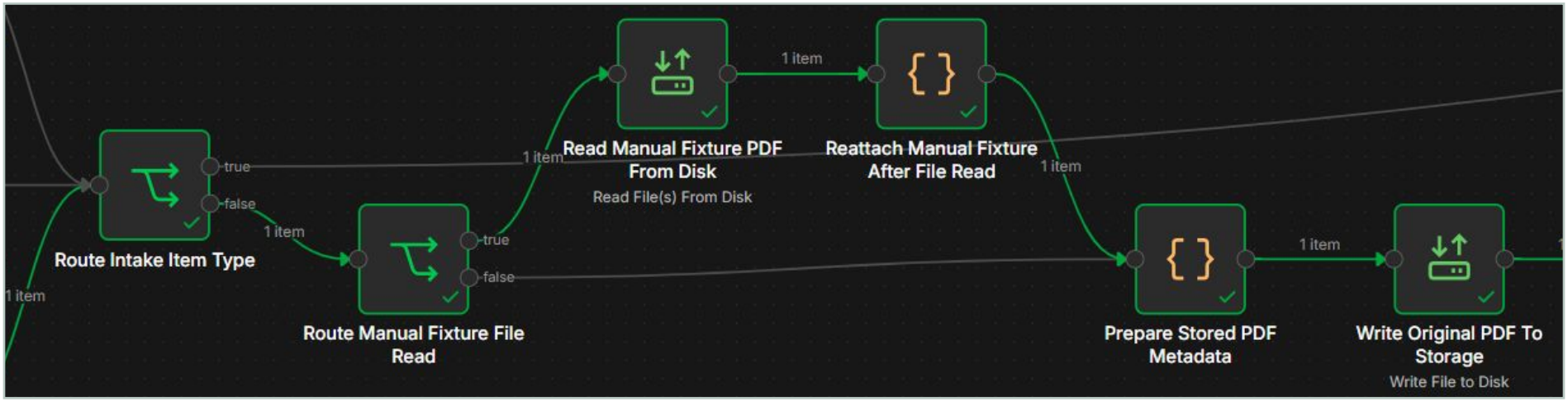
Verified use

The preserved form run created certificates 1, 2 and 3. The fresh October run used the manual test trigger through the same storage, extraction and validation pipeline.

01 / INTAKE STORAGE

Keep the original with its context

Enlarged storage branch from the fresh Apex run; green connections show the executed path.



Input to storage

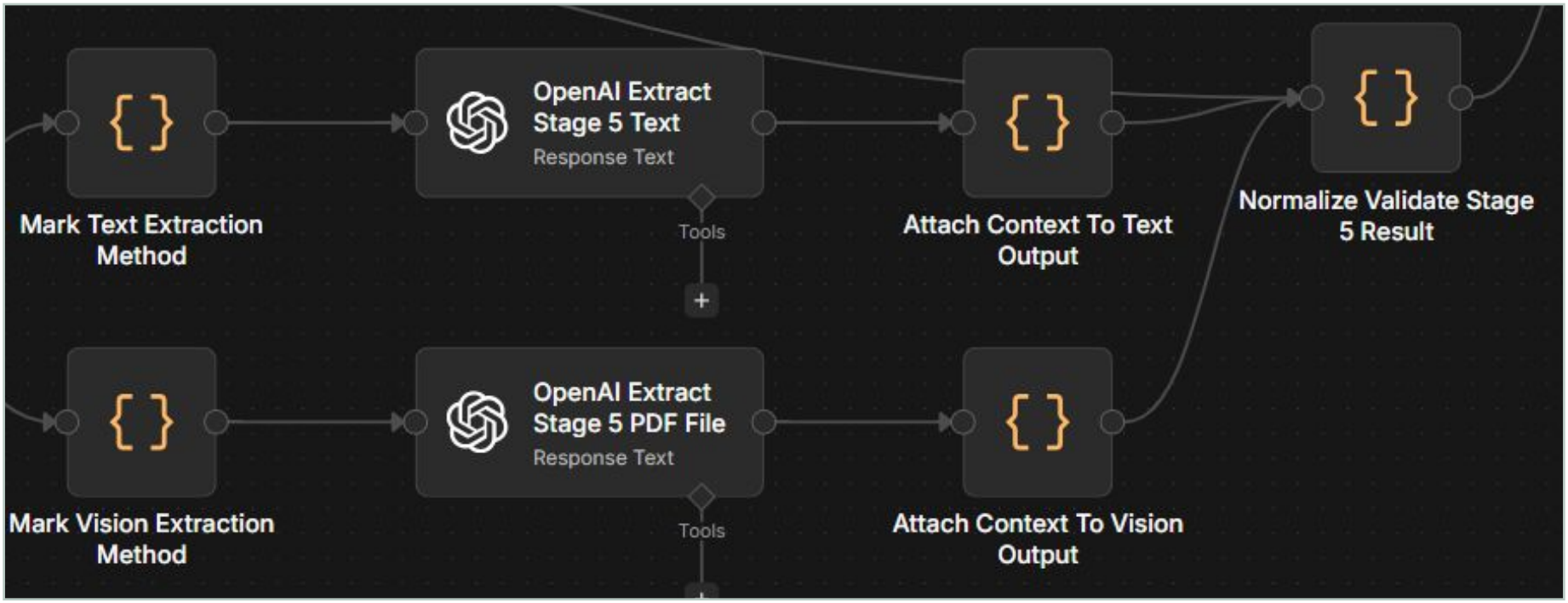
The manual demonstration loads one synthetic PDF. Uploaded and emailed PDFs arrive with their attachment data already present.

Persistent source document

Metadata is prepared and the original file is written to storage. Context is reattached so extraction retains the vendor and source relationship.

Extract fields, then apply explicit rules

Actual OpenAI text and PDF branches converge on validation.



Choose an extraction path

Text-rich PDFs use extracted text; low-text documents use the PDF/file path. Password-protected documents are routed to review before extraction.

Validation is separate from AI

The model transcribes fields; code checks dates, holder, limits and vendor matching. Filing a document does not independently verify insurance or endorsements.

01 / PRESERVED FORM EXECUTION

Three uploads, three stored outcomes

Actual database-node output from the August 2026 form run. Source references are redacted.

Apex Plumbing Services LLC

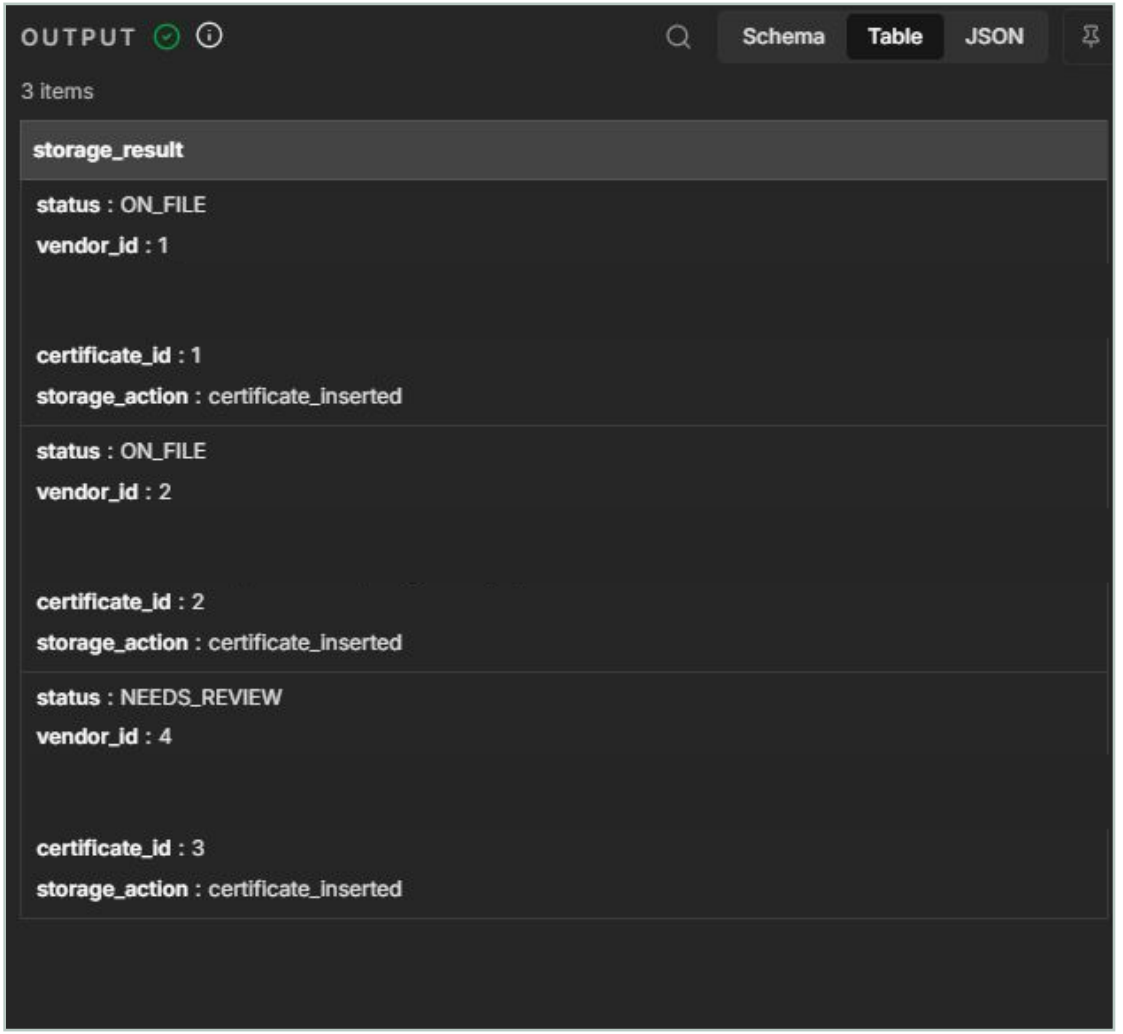
Vendor 1 → certificate 1, ON_FILE.

Blue Ridge HVAC Co.

Vendor 2 → certificate 2, ON_FILE. Expiry dates are tracked separately from filing status.

Summit Shield Roofing Inc.

Vendor 4 → certificate 3, NEEDS_REVIEW. Its inverted dates remain visible on the live register.



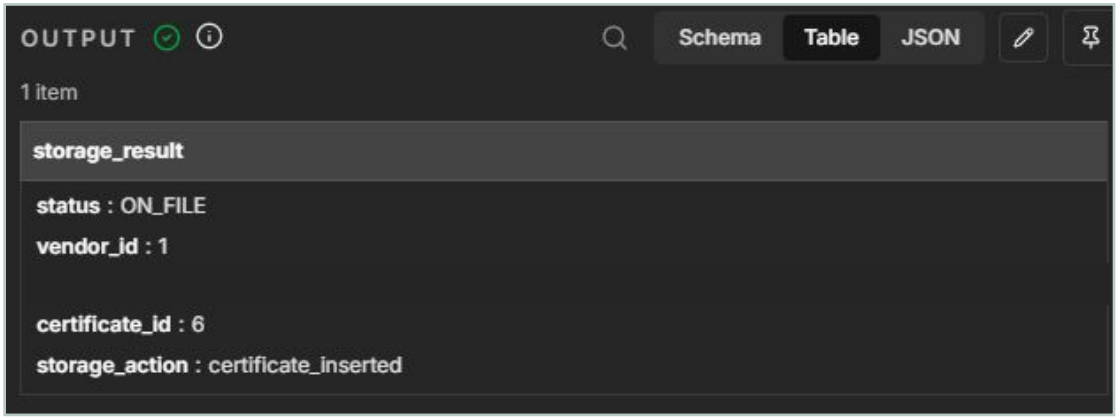
Actual execution output; private fields cropped or redacted.

Apex is stored as certificate 6

Fresh run on 10 October 2026, using one synthetic PDF and real OpenAI extraction.

Input

One synthetic Apex Plumbing Services LLC PDF entered through the manual test trigger.



Executed steps

The run saved the original, extracted fields with OpenAI, validated the result and wrote a certificate record.

Observable result

Postgres returned ON_FILE for vendor 1, certificate 6, with certificate_inserted. The register shows the same vendor and 30 March 2027 expiry.

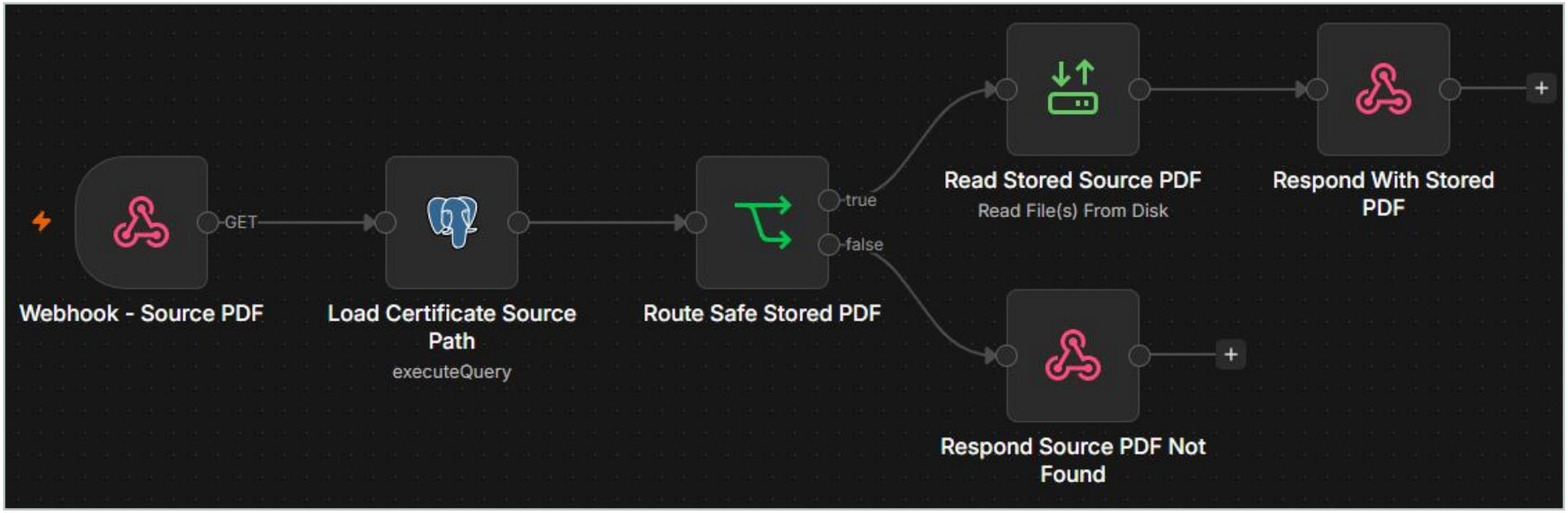


Lower image: vendor details in the actual delivered PDF. The enlarged document is on page 11.

02 / SOURCE DOCUMENT ROUTE

Open the document behind the row

Full actual canvas after Tidy Up. The dashboard passes the certificate record ID.



Request and decision

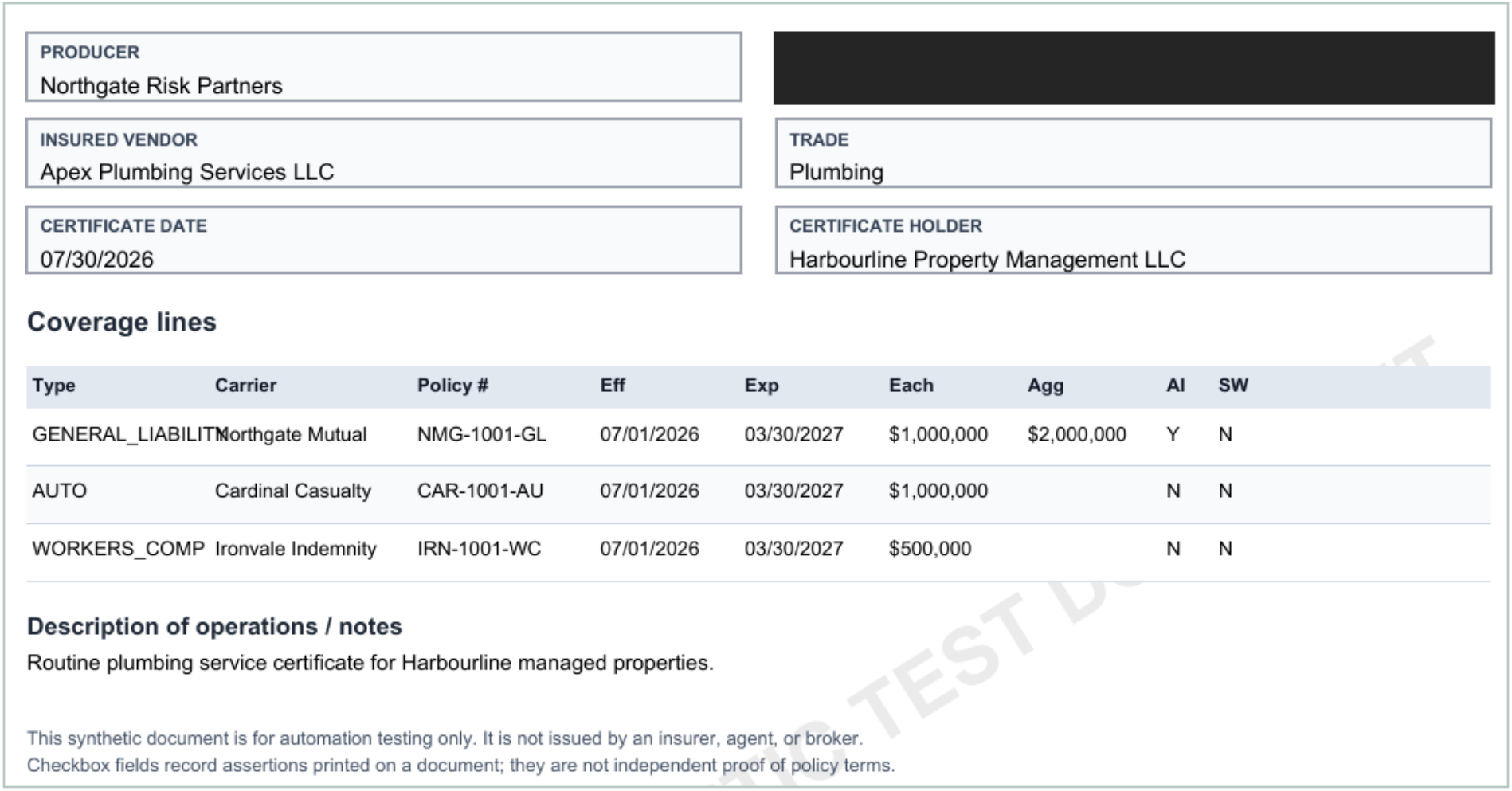
The webhook receives a certificate ID. Postgres returns the stored source path; a safety check decides whether it can be read.

Document response

The successful branch returns the original PDF. A missing or unsafe source uses a not-found response. The route does not decide insurance acceptability.

The source route returned the stored PDF

Certificate 6: HTTP 200, application/pdf, with bytes matching the file written by intake.



PRODUCER
Northgate Risk Partners

[redacted]

INSURED VENDOR
Apex Plumbing Services LLC

TRADE
Plumbing

CERTIFICATE DATE
07/30/2026

CERTIFICATE HOLDER
Harbourline Property Management LLC

Coverage lines

Type	Carrier	Policy #	Eff	Exp	Each	Agg	AI	SW
GENERAL_LIABILITY	Northgate Mutual	NMG-1001-GL	07/01/2026	03/30/2027	$1,000,000	$2,000,000	Y	N
AUTO	Cardinal Casualty	CAR-1001-AU	07/01/2026	03/30/2027	$1,000,000		N	N
WORKERS_COMP	Ironvale Indemnity	IRN-1001-WC	07/01/2026	03/30/2027	$500,000		N	N

Description of operations / notes

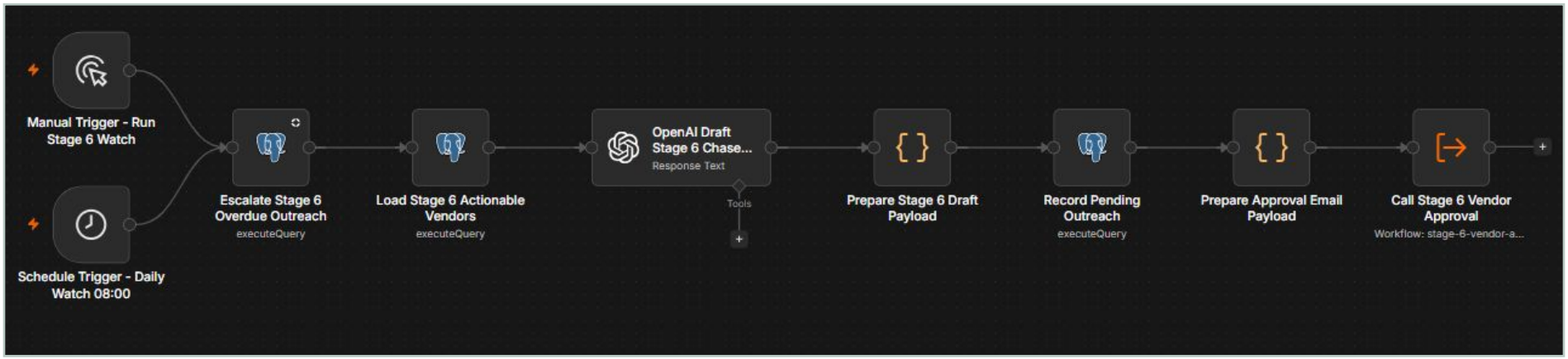
Routine plumbing service certificate for Harbourline managed properties.

This synthetic document is for automation testing only. It is not issued by an insurer, agent, or broker.
Checkbox fields record assertions printed on a document; they are not independent proof of policy terms.

The delivered document names Apex Plumbing Services LLC and the expected certificate holder. Its coverage dates match the register. Producer contact details are flattened out of this image.

Turn the register into an action queue

Full actual canvas after Tidy Up. Daily and manual triggers share the pipeline.



Incoming state

The workflow reads vendors, certificate dates and outreach history from Postgres. It handles overdue outreach before loading actionable vendors.

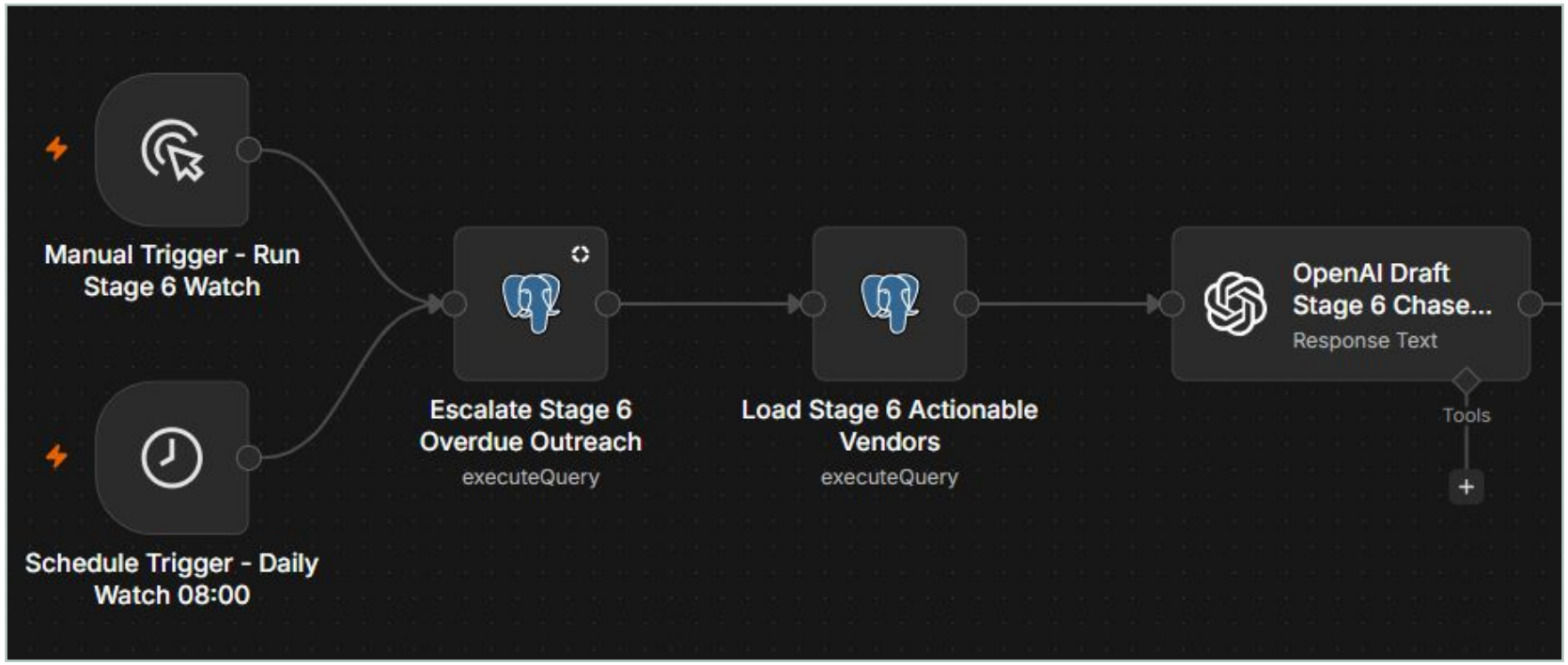
Result and approval handoff

OpenAI drafts requests; pending outreach is recorded and a separate approval execution is called for each vendor. A person controls the send decision.

03 / DAILY WATCH SELECTION

Select vendors that need attention

Enlarged trigger, escalation and selection steps from the real workflow.



Trigger

The schedule is set for 08:00. The preserved proof run was started manually, making its selection and output available for review.

Decisions in durable state

Database functions apply expiry buckets, suppression and follow-up rules. The preserved August run returned 17 actionable vendors.

Record each request before approval

Enlarged handoff branch, captured from the actual n8n editor.



Draft to a pending row

The drafted subject and body are attached to vendor context. Pending outreach is recorded before preparing the approval email.

One approval execution per vendor

Each vendor receives its own workflow execution, giving every request an independent wait and a separate sent or declined result.

03 / ACTUAL QUEUE OUTPUT

Seventeen requests recorded as pending

Preserved August run: actual Postgres output, with the 17-item count and the first stored rows.

Observable business output

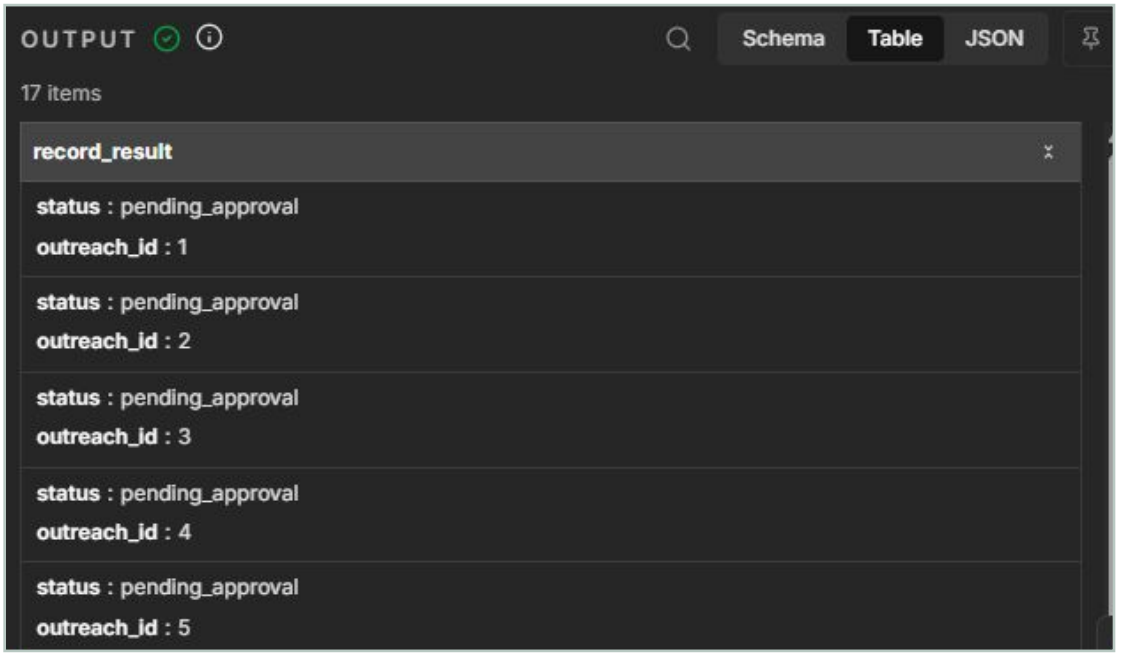
The workflow created 17 pending approval records.

Follow one request

Northstar Snow Removal LLC was outreach 10. Its approval, Gmail response and persisted sent status appear on page 18.

Demonstration boundary

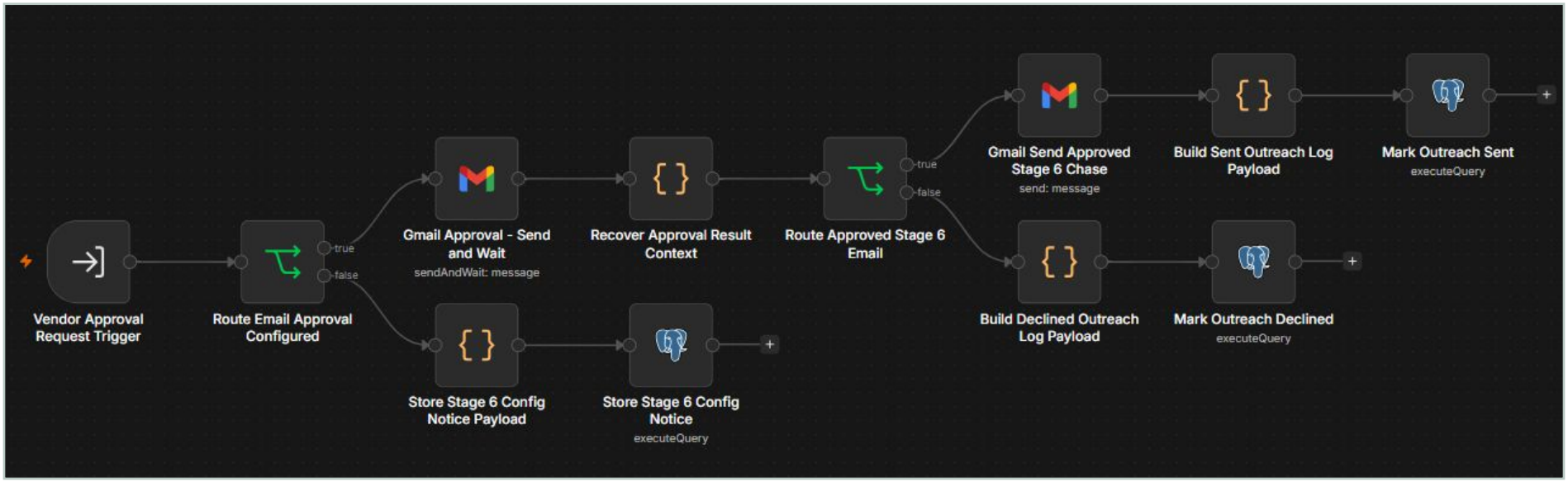
This run used synthetic vendors, real OpenAI drafting and a test mailbox for Gmail delivery.



Actual execution output; private fields cropped or redacted.

A human decision gates every send

Full actual canvas after Tidy Up. The caller provides one vendor and one drafted request.



Incoming request

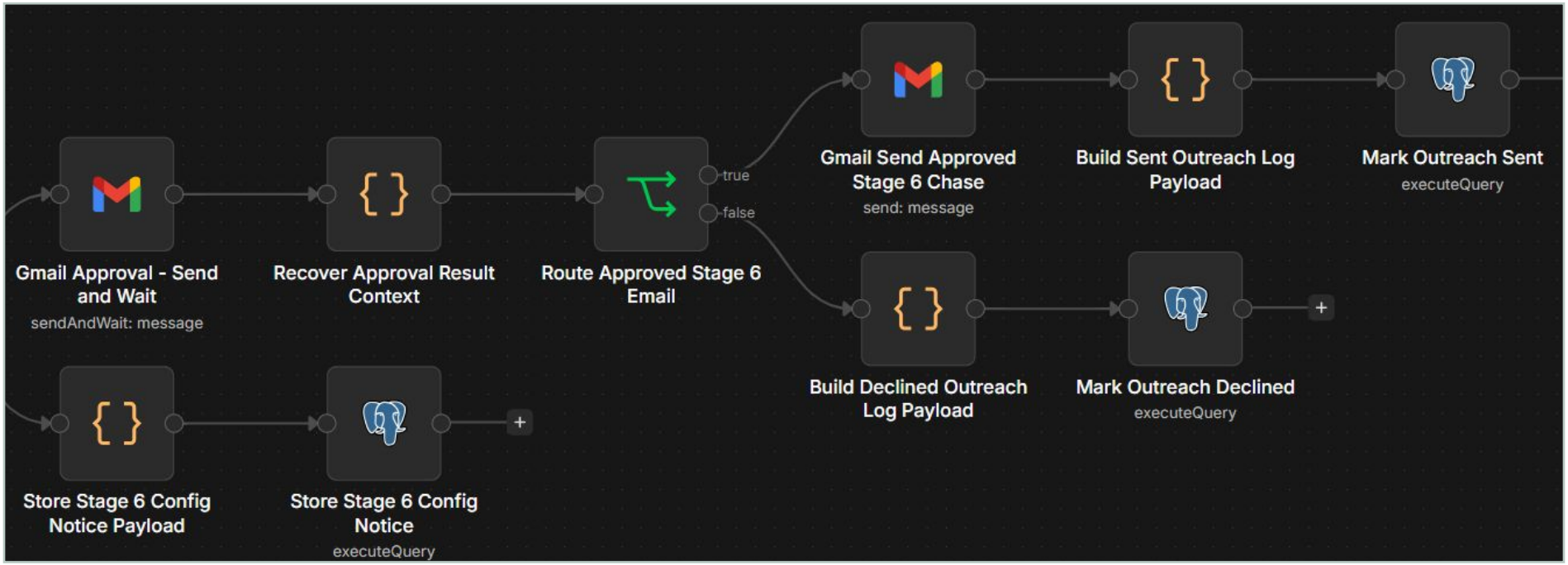
Approval configuration is checked. Gmail sends an approval request and waits for a response in that vendor's own execution.

Resulting state

Approved requests use Gmail send and record sent status in Postgres. False approval responses record declined status. Configuration issues follow a notice branch.

The decision controls send or stop

Enlarged decision branch with readable node labels.



Approved branch

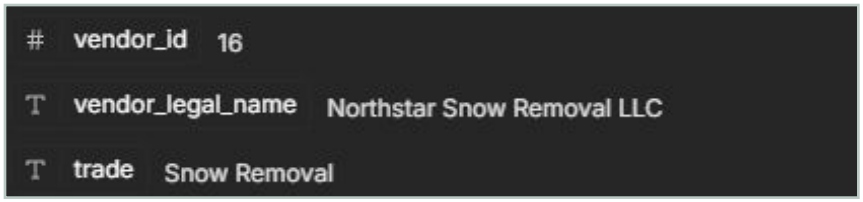
Gmail sends the prepared request. Its response is carried into the outreach log, including the follow-up interval.

Controlled stop

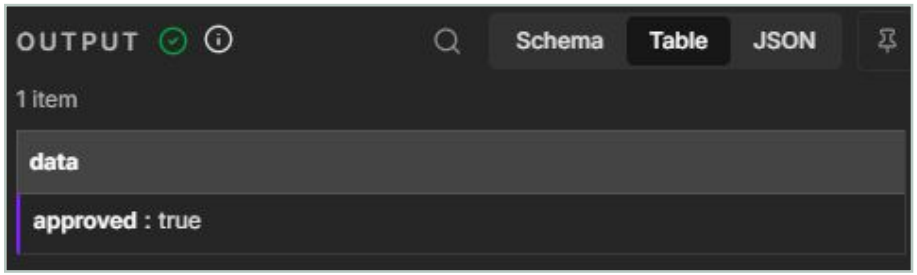
The false branch records a declined request. It does not pass through the Gmail send node. Every outcome belongs to an independent vendor execution.

Northstar: approved, sent and recorded

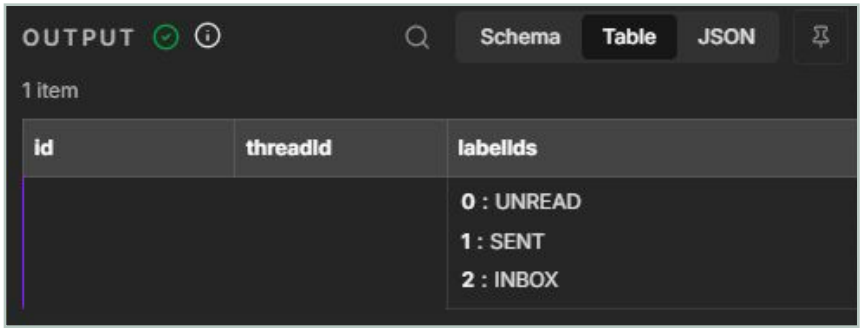
Preserved August run for synthetic Northstar Snow Removal LLC; real Gmail used a test recipient.



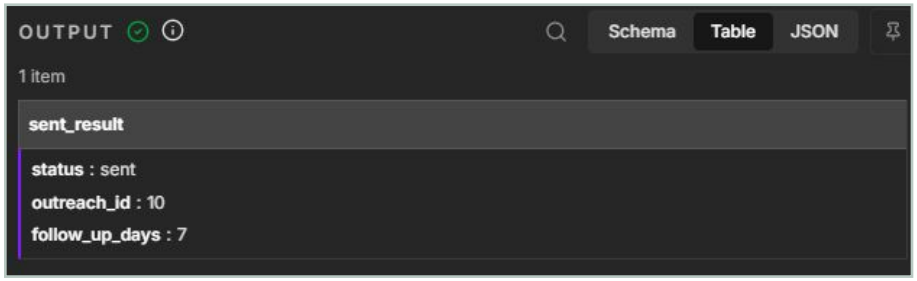
1 / Synthetic vendor context



2 / Gmail approval response: true



3 / Real Gmail response includes SENT



4 / Postgres records outreach 10 as sent

The same execution connects Northstar's approval to Gmail delivery and a durable sent result with a seven-day follow-up interval. Contacts and Gmail message identifiers are removed from these captures.

Vista: request recorded as declined

Preserved August run for synthetic Vista Glass and Door LLC, outreach 15.

Observable stop result

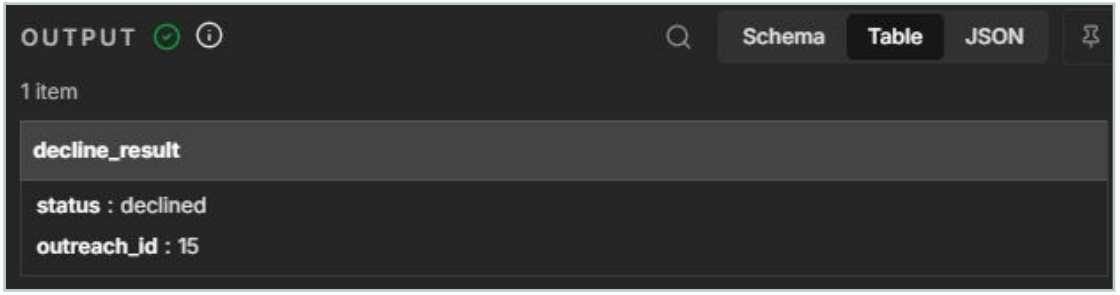
The database node returned status: declined and outreach_id: 15.

What this execution establishes

The stored response was false and the declined branch ran. This execution contains no Gmail send operation.

Separate vendor journey

Vista's result stays separate from Northstar's approved outreach 10.



Actual execution output; private fields cropped or redacted.

05 / NON-PDF REPLY NOTICE

Keep messages without a PDF visible

Full actual workflow after Tidy Up. Gmail messages are normalized into stored notices.



Trigger and incoming data

Gmail supplies received messages. The normalizer skips messages with a PDF attachment; those documents belong to intake.

Output and intervention

Messages without PDFs become intake notices for manual follow-up. The workflow does not create a certificate from an email body.

Ten messages became durable notices

Preserved August test-mailbox run: actual database results, with 10 items and notice IDs.

What this run exercised

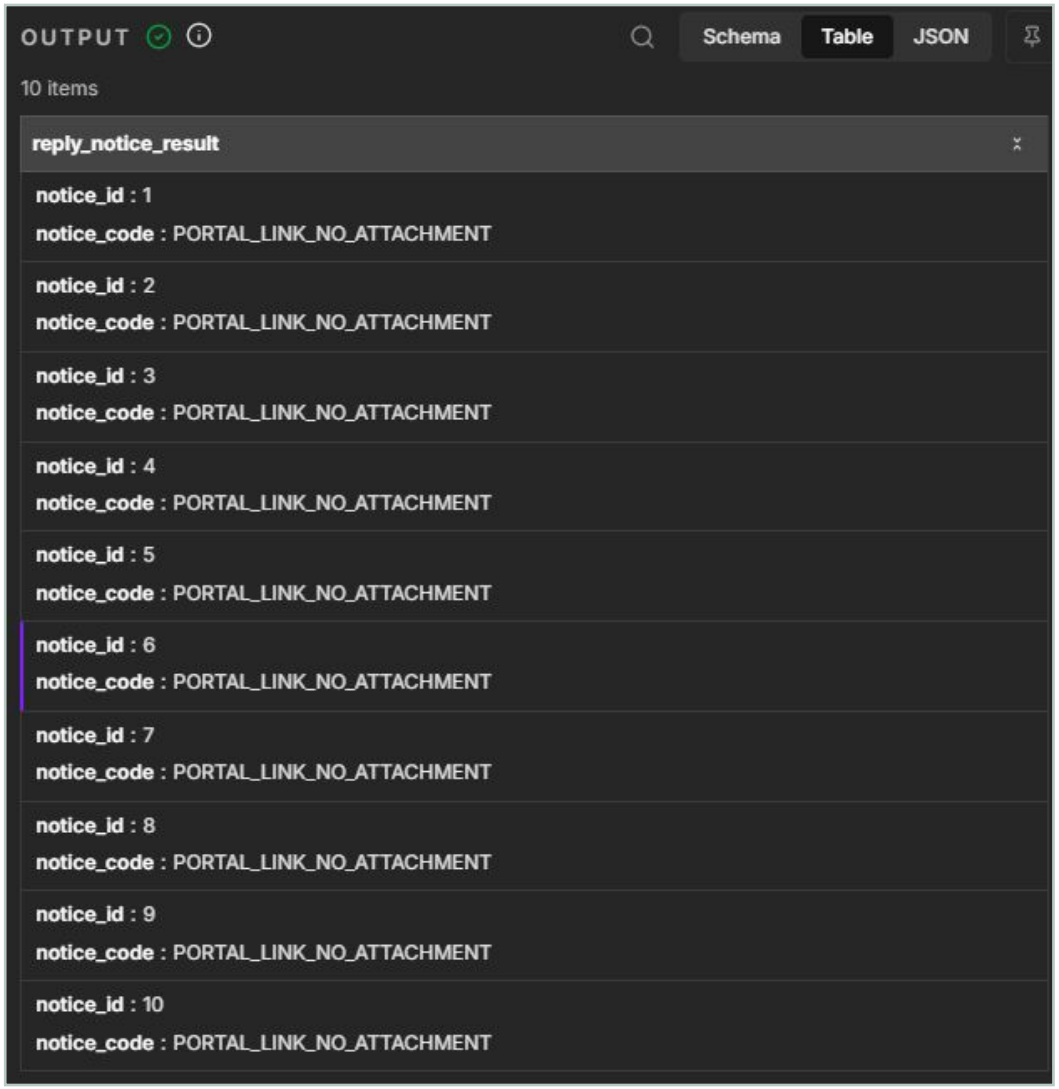
Approval messages received by the test mailbox had no PDF attachments and passed through the notice capture path.

Result in the application

Notice IDs 1–10 returned PORTAL_LINK_NO_ATTACHMENT. This code labels a notice, not a filed certificate.

Business purpose

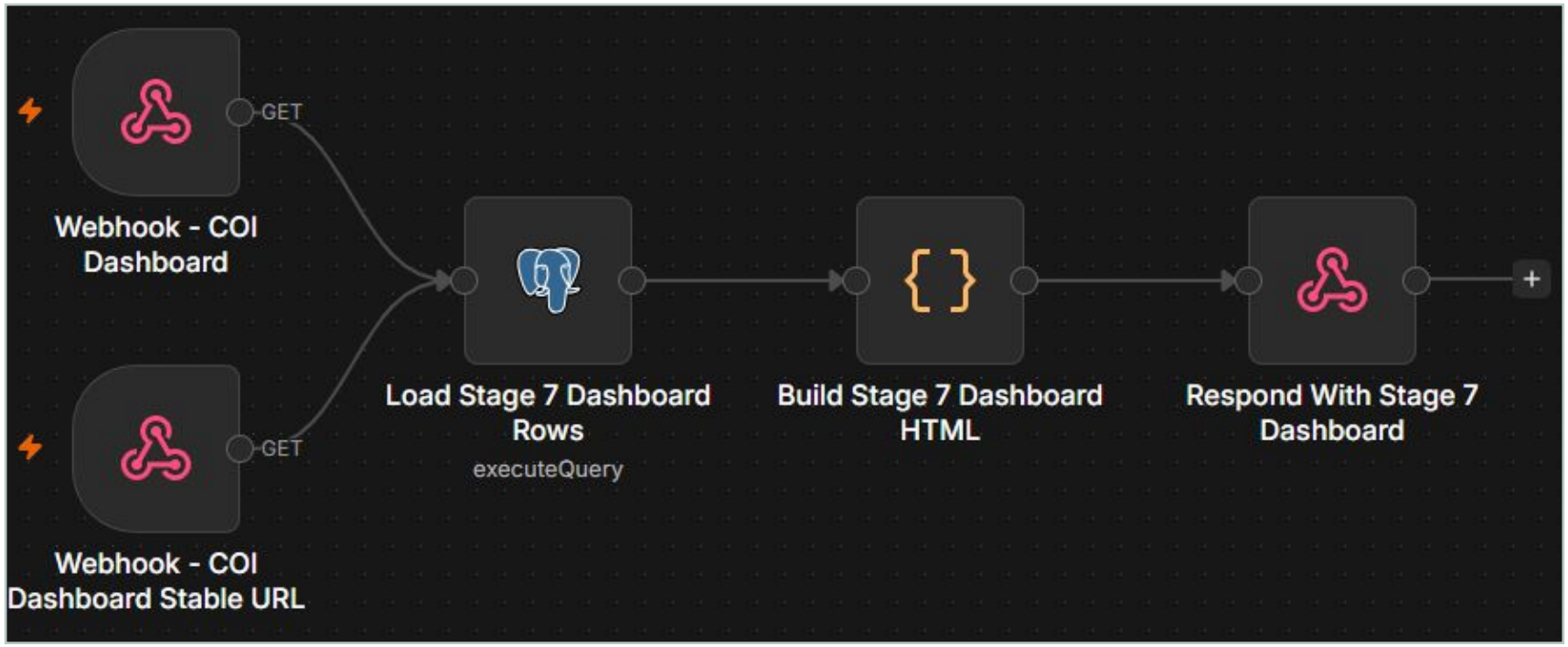
A person can follow up on messages that did not deliver a usable PDF, while document intake remains a separate process.



Actual execution output; private fields cropped or redacted.

Make stored state easy to act on

Full actual workflow after Tidy Up. A browser request triggers a fresh database read.



Request to register

The webhook loads vendor rows from Postgres, builds the register HTML and responds to the browser.

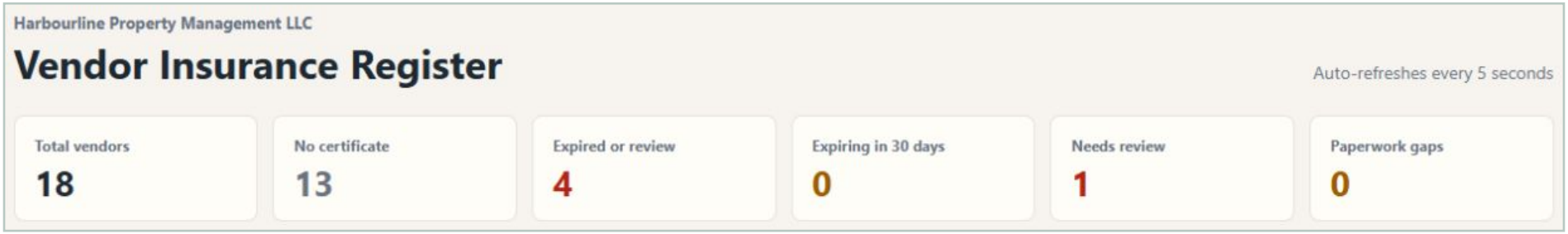
What a person sees

Counts, dates, document flags, chase history and source links appear together. The displayed page refreshes every five seconds.

06 / ACTUAL DASHBOARD OUTPUT

Apex's record is visible in the register

Actual dashboard served by n8n on 10 October 2026. Every displayed vendor is synthetic.



VENDOR	TRADE	CARRIER	POLICY #	GL LIMIT	EXPIRES	STATUS	CHASE
Apex Plumbing Services LLC	Plumbing	Northgate Mutual	NMG-1001-GL	$1M	2027-03-30T00:00:00.000Z	Current until Mar 30, 2027	Not yet chased
Summit Shield Roofing Inc. date inverted · no wc line · no cancellation notice wording	Roofing	Northgate Mutual	NMG-5005-GL	$2M	2026-08-01T00:00:00.000Z	Needs review	Not yet chased

Apex appears current until 30 March 2027; Summit Shield is marked Needs review. The register reads actual stored records. Certificate 6's source link returned the original PDF successfully.

VERIFIED DEMONSTRATION

What this build demonstrated

Evidence connects inputs and decisions to stored records, delivered files and outreach outcomes.

Document intake	Fresh Apex PDF → certificate 6 → current register row → original PDF returned.
Form submission	Three submitted PDFs → three stored results, including a document marked for review.
Daily watch	17 selected vendors → 17 durable pending approval records.
Approved outreach	Northstar approval true → Gmail SENT response → outreach 10 recorded as sent.
Controlled stop	Vista false response → outreach 15 declined; send node not executed.
Message capture	10 test-mailbox messages without PDFs → 10 durable notices.

Documents become traceable records, dates become actionable requests and human decisions become recorded outreach outcomes.